Jiawei Lian, Curriculum Vitae

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Research Gate Github Homepage

-**Birth Date**: 01/1997
-**Status**: Third-year Ph.D. student, PolyU and NWPU joint training leading to dual Ph.D.
-**Research Interest**: AI Safety, Remote sensing, Computer vision, Large model
-**Postal Address**: Room 505, Building CF, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong

Education

01/2024 – Present	Ph.D., Artificial Intelligence and Signal Processing, Advisor: Lap-Pui Chau The Hong Kong Polytechnic University (PolyU) Thesis: *Towards Exploring the Adversarial Robustness of Deep Learning Models*
04/2022 – Present	Ph.D., Information and Communication Engineering, Advisor: Shaohui Mei Northwestern Polytechnical University (NWPU) Thesis: *Research on Adversarial Attacks against Object Detection Models*
09/2019 – 03/2022	M.E., Control Engineering, Advisor: Junhong He Northwestern Polytechnical University Thesis: *Research on Rapid Detection of Surface Defects Based on Deep Learning*
09/2015 – 07/2019	B.E., Automation, Advisor: Junlin Zhu JiangXi University of Science and Technology Thesis: *Design of 4R Joint Robot Control System based on PROFIBUS and T-CPU*

Research Publications (Please refer to Google Scholar for full list)

Journal Articles

1. J. Lian, S. Mei, X. Wang, *et al.*, "Attack anything: Blind dnns via universal background adversarial attack," *arXiv preprint arXiv:2409.00029*, 2024. (**Under review, IEEE Transactions on Dependable and Secure Computing**).
2. J. Lian, J. Pan, L. Wang, Y. Wang, S. Mei, and L.-P. Chau, "Padetbench: Towards benchmarking physical attacks against object detection," *arXiv preprint arXiv:2408.09181*, 2024. **(Under review, Knowledge-Based Systems)**.
3. J. Lian, J. Pan, L. Wang, Y. Wang, S. Mei, and L.-P. Chau, "Revealing the intrinsic ethical vulnerability of aligned large language models," *arXiv preprint arXiv:2504.05050*, 2025. (**Under review, Nature Communications**).
4. J. Lian, J. Pan, L. Wang, Y. Wang, S. Mei, and L.-P. Chau, "Semantic representation attack against aligned large language models," 2025. (**Under review, NeurIPS 2025**).
5. S. Mei, J. Lian, X. Wang, Y. Su, M. Ma, and L.-P. Chau, "A comprehensive study on the robustness of image classification and object detection in remote sensing: Surveying and benchmarking," *Journal of Remote Sensing*, 2023. (**The first author is the Ph.D. advisor**).
6. J. Lian, X. Wang, Y. Su, M. Ma, and S. Mei, "CBA: Contextual background attack against optical aerial detection in the physical world," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1–16, 2023.
7. J. Lian, S. Mei, S. Zhang, and M. Ma, "Benchmarking adversarial patch against aerial detection," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–16, 2022.

8. J. Lian, J. He, Y. Niu, and T. Wang, "Fast and accurate detection of surface defect based on improved yolov4," *Assembly Automation*, vol. 42, no. 1, pp. 134–146, 2021.
9. J. Lian, J. He, Y. Niu, and T. Wang, "Rapid detection of surface defects based on multi-scale compression cnn," *Computer Integrated Manufacturing System*, vol. 28, no. 11, p. 3624, 2021.

Conference Articles

1. J. Lian, X. Wang, Y. Su, M. Ma, and S. Mei, "Contextual adversarial attack against aerial detection in the physical world," in *IGARSS 2023-2023 IEEE International Geoscience and Remote Sensing Symposium*, IEEE, 2023, pp. 6632–6635.

China Patents

- 16/05/2023 — A method and device for generating adversarial camouflage patterns against object detection. (Patent acceptance number: 202310547602.8)
- 16/12/2022 — A method and device for physical attack-oriented self-adaptive adversarial patch generation. (Patent acceptance number: 202211617496.8)
- 15/12/2022 — A method and device for generating adversarial patches for unobstructed physical adversarial attacks. (Patent acceptance number: 202211612346.8)
- 05/06/2021 — A fast surface defect detection method based on the lightweight convolutional neural network. (Patent acceptance number: 202110627893.2)

Miscellaneous Experience

Certifications

- 04/2022 — **International English Language Testing System (IELTS): Overall band score 7.0 (Listening 7.5, Reading 7.5, Writing 6.5, Speaking 6.0)**
- 07/2017 — **Fitness Coach Licentiate (FCL): Awarded by The Ministry of Human Resource and Social Security, The People's Republic of China.**

Volunteer Works

- 03/2022 — Volunteer in Northwestern Polytechnical University's COVID-19 epidemic prevention and control volunteer service.
- 07/2019 — Participate in the "Guarding the Qinling Mountains, College Students in Action" public welfare and environmental protection activity.

Involved Projects

- 09/2022-02/2023 — Research on the robustness of remote sensing image classification and object detection model based on deep learning.
- 12/2021-08/2022 — Research on background adversarial perturbations for physical attack. **Relevant research achievements received an interview invitation from the British authoritative magazine New Scientist**.

Internship

- 07/2021-09/2021 — Internship at AVIC Xi'an Flight Automatic Control Research Institute.
- 07/2020-09/2020 — Internship at Xi'an Institute of Modern Control Technology.